

EXPERIMENT – 02

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

PRIMARY KEY:

```
mysql> -- Create the tables used for constraint testing
mysql> create table student1(RegNo int, S1Name varchar(15), Age char(2), DOB int, address varchar(15));
Query OK, 0 rows affected (0.38 sec)

mysql> create table student2(RegNo int, S1Name varchar(15), Age char(2), MobileNo int, address varchar(15));
Query OK, 0 rows affected (0.04 sec)

mysql> create table student3(RegNo int, S1Name varchar(15), Age char(2), MobileNo int, address varchar(15));
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> -- 1) PRIMARY KEY on student2
mysql> alter table student2 add primary key(RegNo);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | NO   | PRI | NULL    |       |
| S1Name | varchar(15)   | YES  |     | NULL    |       |
| Age    | char(2)       | YES  |     | NULL    |       |
| MobileNo | int         | YES  |     | NULL    |       |
| address | varchar(15)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.05 sec)
```

FOREIGN KEY:

```
mysql> -- 2) FOREIGN KEY test: student3.RegNo references student2.RegNo
mysql> alter table student3 add foreign key(RegNo) references student2(RegNo);
Query OK, 0 rows affected (0.15 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student3;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	YES	MUL	NULL	
S1Name	varchar(15)	YES		NULL	
Age	char(2)	YES		NULL	
MobileNo	int	YES		NULL	
address	varchar(15)	YES		NULL	

```
5 rows in set (0.01 sec)
```

```
mysql>
```

```
mysql> -- 3) PRIMARY KEY + CHECK on faculty
```

```
mysql> alter table faculty add primary key(Facno);
```

```
Query OK, 0 rows affected (0.07 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> alter table faculty add check(gender='M' or gender='F');
```

```
Query OK, 0 rows affected (0.09 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc faculty;
```

Field	Type	Null	Key	Default	Extra
Facno	int	NO	PRI	NULL	
FacName	char(15)	YES		NULL	
gender	char(1)	YES		NULL	
Dob	int	YES		NULL	
mobilenno	int	YES		NULL	
DOJ	int	YES		NULL	

```
6 rows in set (0.00 sec)
```

UNIQUE & CHECK:

```
mysql>
mysql> -- 3) PRIMARY KEY + CHECK on faculty
mysql> alter table faculty add primary key(Facno);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table faculty add check(gender='M' or gender='F');
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | int | NO | PRI | NULL | |
| FacName | char(15) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| Dob | int | YES | | NULL | |
| mobileno | int | YES | | NULL | |
| DOJ | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
mysql> -- 4) PRIMARY KEY on department
mysql> alter table department add primary key(deptno);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| deptno | int | NO | PRI | NULL | |
| deptname | char(10) | YES | | NULL | |
| depthead | char(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> -- 6) UNIQUE KEY on student2.S1Name
mysql> alter table student2 add unique key(S1Name);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int | NO | PRI | NULL | |
| S1Name | varchar(15) | YES | UNI | NULL | |
| Age | char(2) | YES | | NULL | |
| MobileNo | int | YES | | NULL | |
| address | varchar(15) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> -- 7) Modify student1.address to NOT NULL
mysql> alter table student1 modify address int not null;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student1;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int | YES | | NULL | |
| S1Name | varchar(15) | YES | | NULL | |
| Age | char(2) | YES | | NULL | |
| DOB | int | YES | | NULL | |
| address | int | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```